Box 1: The BAD multitrait biodiversity model

We outline a model to evaluate the effect of trait architecture on biodiversity. Specifically, we implement a trait-based metacommunity model to explore how interactions among biotic, abiotic and dispersal traits affects the formation of hot and cold spots in biodiversity. We explore two architectures rooted in empirical and theoretical studies (refs). The modular and the magic trait architecture. The modular approach takes into account independent traits suggesting each trait contributes independently to fitness (Figure 1a:Modular matrix). The magic trait approach incorporates the biotic trait as the main contributor to fitness with the abiotic and the dispersal traits following the biotic trait (Fig 1b:Magic matrix).

Consider many resource and consumer populations characterized each by individuals containing three normally distributed traits (biotic, z_b, abiotic, z_a, and dispersal, z_d, represented as $\mathbf{z} = [z_b,\ z_a,\ z_d]$ for phenotype i). Populations are located in a network of discrete sites connected by migration events and the local population demography is driven by the temporal dependent fitness function accounting for trait architecture following

$$\mathrm{W}(\mathbf{z})_x^t = exp[-\gamma(((\mathbf{z}_x^t - \theta_x)^2)^T \omega^{-1}(\mathbf{z}_x^t - \theta_\mathbf{x})^\mathbf{2})] \, , \tag{1}$$

where $\mathbf{z}_x^t$ is the vector of trait values of phenotype i at time t and site x, θ_x is the multivariate fitness optimum, ω is the covariance matrix (Fig 1) (Lande, 1980; Melo and Marroig, 2014), and γ determines the interaction sensitivity to deviations from the biotic (i.e., coevolutionary selection strength), abiotic and dispersal optimum (Q1: W function for resources and consumers). If the covariance matrix, ω, is diagonal, representing our modular scenario (Fig. 1), then there is no correlated stabilizing selection. If the covariance matrix is not diagonal and considering our magic scenario (Fig. 1), then there is correlated stabilizing selection with the biotic trait determining the selection on the other traits. The biotic optimum changes in time and we define it as the minimum and maximum mean trait distance across all populations in site x for the consumer and the resource biotic trait value, respectively. The dispersal optimum is given by the mean distance in the landscape (i.e., each site has coordinates from which we calculate the Euclidian distance between each pair of sites). Phenotypes with dispersal trait value near to this optimum maximize fitness for this trait only for the modular but not for the magic trait scenario. The abiotic optimum is taken initially as the mean of the distribution and it remains the same during the dynamics (Q2: moving BAD optimum?)

The time dependent fitness function in site x, $\mathrm{W}(\mathbf{z})_x^t$, determines the mother of the offspring. To obtain the phenotypic value for the offspring, $z_o = [z_{ob}, z_{oa}, z_{od}]$, we add to each trait value of the mother the environmental deviation, $\mathbf{e}$, taken from a multivariate uniform distribution with range ± 0.001 in all dimensions (Q3: Take into account additive effects and pleiotropy matrix?). We run our model G generations each following the selection, mating and dispersion scheme to produce a new generation per site. We explore a gradient of dispersal and coevolutionary selection strength, to investigate the formation of hot and cold spots formation in local and regional biodiversity (Fig. 2.)

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